BDT301 — Group Project Rubric

End-to-End Big Data Analytics Pipeline | Total: 100 points

No.	Criterion	Points	CLO
1	Problem definition & dataset	10	CLO1, CLO2
2	Architecture & technology justification	15	CLO1–CLO5
3	Data ingestion & storage	15	CLO2, CLO3
4	Spark processing	20	CLO4
5	Pipeline integration	15	CLO1–CLO5
6	Analysis & insight	10	CLO1, CLO5
7	Reliability, data quality & trade-offs	5	CLO2–CLO5

8	Documentation & reproducibility	5	CLO1–CLO5
9	Presentation, demo & Q&A	5	CLO1–CLO5
	Total	**100**	

Excellent	Good
9–10: Problem, users, and ≥3 analytical questions are clear. Dataset is ≥1 GB, relevant, documented, and quality issues are identified.	7–8: Problem and questions are clear. Dataset meets size requirement and is documented, with minor gaps.
13–15: Complete end-to-end architecture, all major components are justified using workload, scalability, reliability, and maintainability trade-offs.	10–12: Architecture is complete and mostly justified, trade-off discussion is present but limited.
13–15: Repeatable ingestion works on the full dataset, raw and processed data are separated, storage design is appropriate and documented.	10–12: Ingestion and storage work with minor issues, design is mostly appropriate and reproducible.
17–20: Spark pipeline runs on the project dataset, meaningful cleaning/transformation and ≥2 analytical aggregations are implemented, execution evidence and reasoning are clear.	13–16: Spark processing works and supports the business questions, minor technical or analytical limitations remain.
13–15: Data flows through all required stages from source to analytical output, run order is clear and the main path is reproducible.	10–12: Main pipeline works with small manual steps or minor integration issues.
9–10: Results answer all stated business questions and are interpreted correctly with useful evidence.	7–8: Most questions are answered and results are interpreted with minor gaps.
5: At least 2 realistic failure/quality/scalability concerns are analyzed, with practical mitigations and at least 1 clear architecture trade-off.	4: Concerns and trade-offs are identified with reasonable explanation.

5: README, architecture diagram, dataset notes, commands, and report allow another person to reproduce the main pipeline.	4: Documentation is complete with only minor missing steps.
5: Demo is clear and working, all members explain their responsibilities and answer technical questions accurately.	4: Demo works with minor issues, most answers show good understanding.

Pass	Insufficient
5–6: Basic problem and dataset are provided, but questions or dataset documentation are incomplete.	0–4: Problem is unclear, dataset is too small/irrelevant, or key dataset evidence is missing.
7–9: Architecture covers the required pipeline but justification is shallow or some components are weakly connected.	0–6: Architecture is incomplete, inconsistent with implementation, or tools are selected without explanation.
7–9: Basic ingestion/storage exists but relies on manual steps or has limited reproducibility.	0–6: Ingestion does not work reliably, storage is unsuitable, or evidence is missing.
9–12: Spark is used but processing is simple, incomplete, or only partially supports the questions.	0–8: Spark job cannot be demonstrated, processes only trivial data, or is replaced by local/non-Spark processing.
7–9: Components work individually but integration is partial or fragile.	0–6: Pipeline is largely disconnected or cannot be demonstrated end-to-end.
5–6: Some analytical output exists but interpretation is weak or only partly linked to the questions.	0–4: Output is mostly screenshots/raw tables with little or no meaningful interpretation.
3: Basic issues are listed but analysis or mitigation is limited.	0–2: Little evidence of reliability, quality, or trade-off analysis.

3: Documentation exists but important setup/run details are missing.	0–2: Documentation is incomplete or the project cannot be reproduced from the submission.
3: Presentation is understandable but demo or Q&A reveals notable gaps.	0–2: Demo fails or members cannot explain submitted work.

Scoring and Individual Adjustment Guide

Section	Rule
Team rubric	Score the 9 criteria out of 100.
Individual understanding	Each member must explain assigned work and overall architecture.
Contribution	Check contribution record against demonstrated work.
AI/tool usage	Use of AI/tools is allowed only when the student can explain and adapt the output.
Minimum project condition	Dataset ≥1 GB and Spark must be used as the main processing engine.

Recommended use	Evidence
Use the rubric descriptors as the primary basis.	Repository, report, demo, outputs.
Use targeted Q&A during final presentation.	Answers, code walkthrough, design explanation.
Ask who implemented and verified each major component.	Git history, task log, files, demo ownership.
Ask the student to explain or modify generated code/diagram when relevant.	Prompt declaration if required by instructor, explanation during Q&A.
Verify before full rubric scoring.	Dataset size evidence and Spark execution evidence.

Notes

Team score before individual adjustment.

A member may receive a lower individual mark if understanding is materially weaker than the team submission.

Do not grade only by claimed percentage.

Do not award credit for output the student cannot explain.

If a mandatory condition is not met, apply the relevant criterion descriptor rather than inventing a separate penalty.

Rubric-to-CLO Mapping

Criterion	CLO1	CLO2	CLO3
Problem definition & dataset	Primary	Primary	
Architecture & technology justification	Primary	Support	Support
Data ingestion & storage		Primary	Primary
Spark processing			
Pipeline integration	Support	Support	Support
Analysis & insight	Primary		
Reliability, data quality & trade-offs		Primary	Support
Documentation & reproducibility	Support	Support	Support
Presentation, demo & Q&A	Support	Support	Support

CLO4 / CLO5

Primary

Support

CLO4 Primary

CLO4/CLO5 Primary

CLO5 Primary

CLO4/CLO5 Support

Support

Support